A 39-DIMENSIONAL ARB KRAWCZYK INTERVAL CERTIFICATE FOR THE THIRD FOURIER FIBRE IN DIMENSION SIX

ARGUS OPEN RESEARCH PROJECT

Abstract. Whether four (or seven) mutually unbiased bases (MUBs) exist in $\mathbb{C}^6$ is a longstanding open problem in quantum information theory. On the real algebraic system for 6×6 complex Hadamard triples, we establish a rigorous 39-dimensional interval-Newton (Krawczyk) inclusion certificate on the third Fourier fibre, proving that the continuation source and the canonical Fourier root lie in a common phase-unambiguous isolating box satisfying all 82 defining polynomial equations and refuting the fixed-source branch separation hypothesis.

1. Main Interval-Newton Certificate

Theorem 1.1. *In the 39-dimensional real gauge-fixed coordinate chart for the third Fourier fibre of the Hadamard-triple system in $\mathbb{C}^6$, the interval Krawczyk operator maps an explicit isolating hyperrectangle strictly into its interior, proving existence and uniqueness of the smooth zero inside that box and certifying that the continuation source coincides with the standard Fourier-family root.*

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